ACEFA
AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

Lecture 6

Analysis of epidemic time-series part 3: quantifying trends in epidemic time series data consisting of signals from multiple pathogens, strains or variants

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Prepared for the ACEFA EpiAnalytics Course at Australian CDC, Canberra, November 2026
Slides prepared by Oliver Eales, Freya Shearer, Alexandra Hogan

This lecture's scope



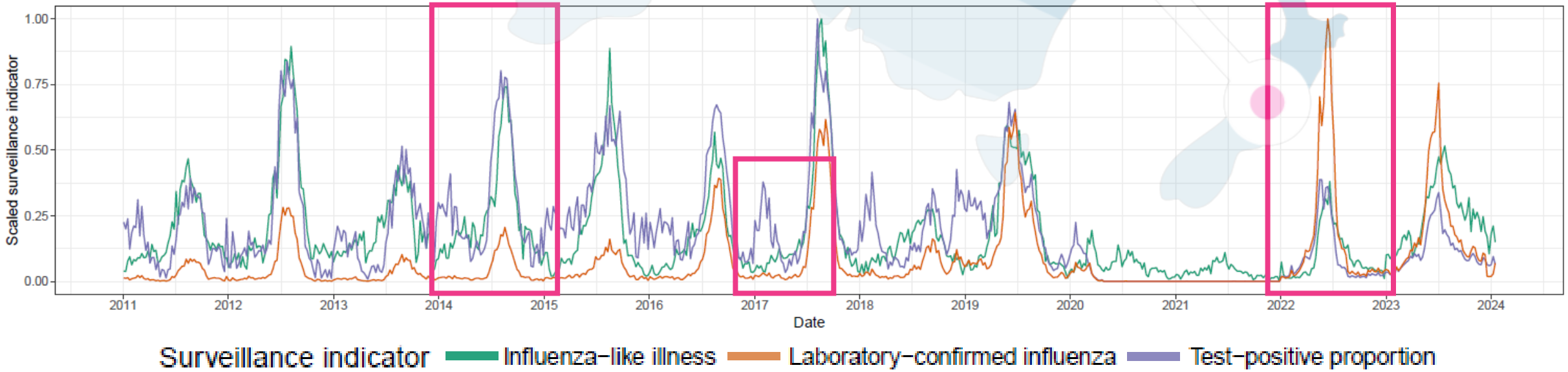
- In previous lectures we:
  - Introduced fundamental theory for analysing time series of infection incidence
  - Discussed biases in commonly available epidemic time series data

- In this lecture we will:
  - Further investigate biases in epidemic time series from routinely collected surveillance data
  - Motivated by a case study of influenza surveillance
  - Discuss methods for improving how we report and analyse influenza surveillance data

Influenza surveillance

Background: why do we conduct influenza surveillance?

- Monitoring the incidence of influenza infections is important for quantifying:
  - transmission dynamics (e.g., peak timing, peak magnitude, time of season onset)
  - clinical severity
- Routine community surveillance typically reports three surveillance indicators:
  - Influenza-like illness (ILI)
  - Laboratory-confirmed influenza
  - The test-positive proportion
- Often substantial differences between the trends of these indicators within and across influenza seasons

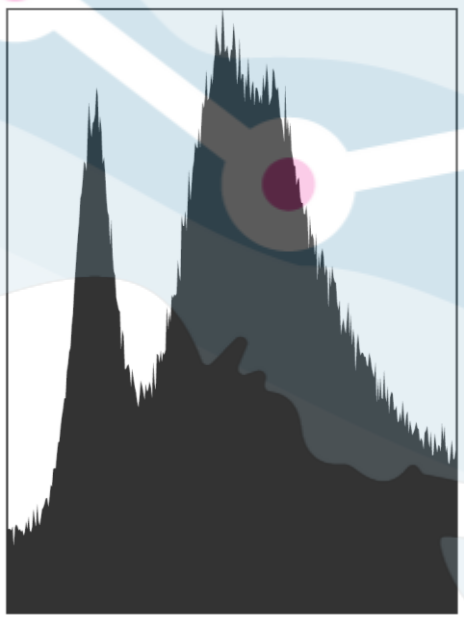


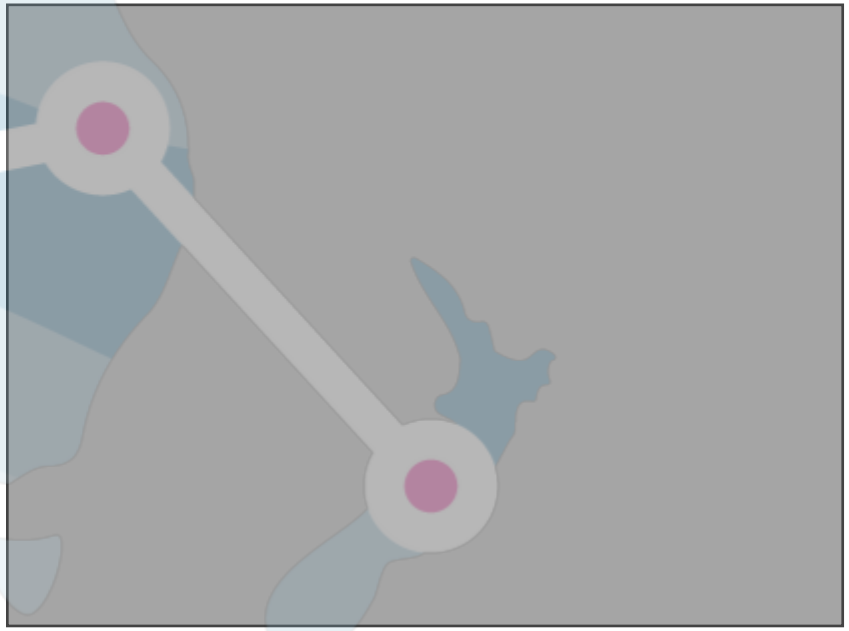
Surveillance indicators: Influenza-like illness (ILI)

- ILI is the rate of influenza-like illness (usually) reported through sentinel healthcare sites per unit time
- Influenza infections represent only a small fraction of ILI
- The ILI time-series regularly used to quantify influenza dynamics mostly reflects the abundance of other pathogens (which also vary over time).

All pathogens | Influenza pathogens | Non-influenza pathogens

$$ILI(a,t) = \sum_{n \in \text{Flu}}^{N} i_n(a,t) P_n(a,t) + \sum_{m \notin \text{Flu}}^{M} i_m(a,t) P_m(a,t)$$

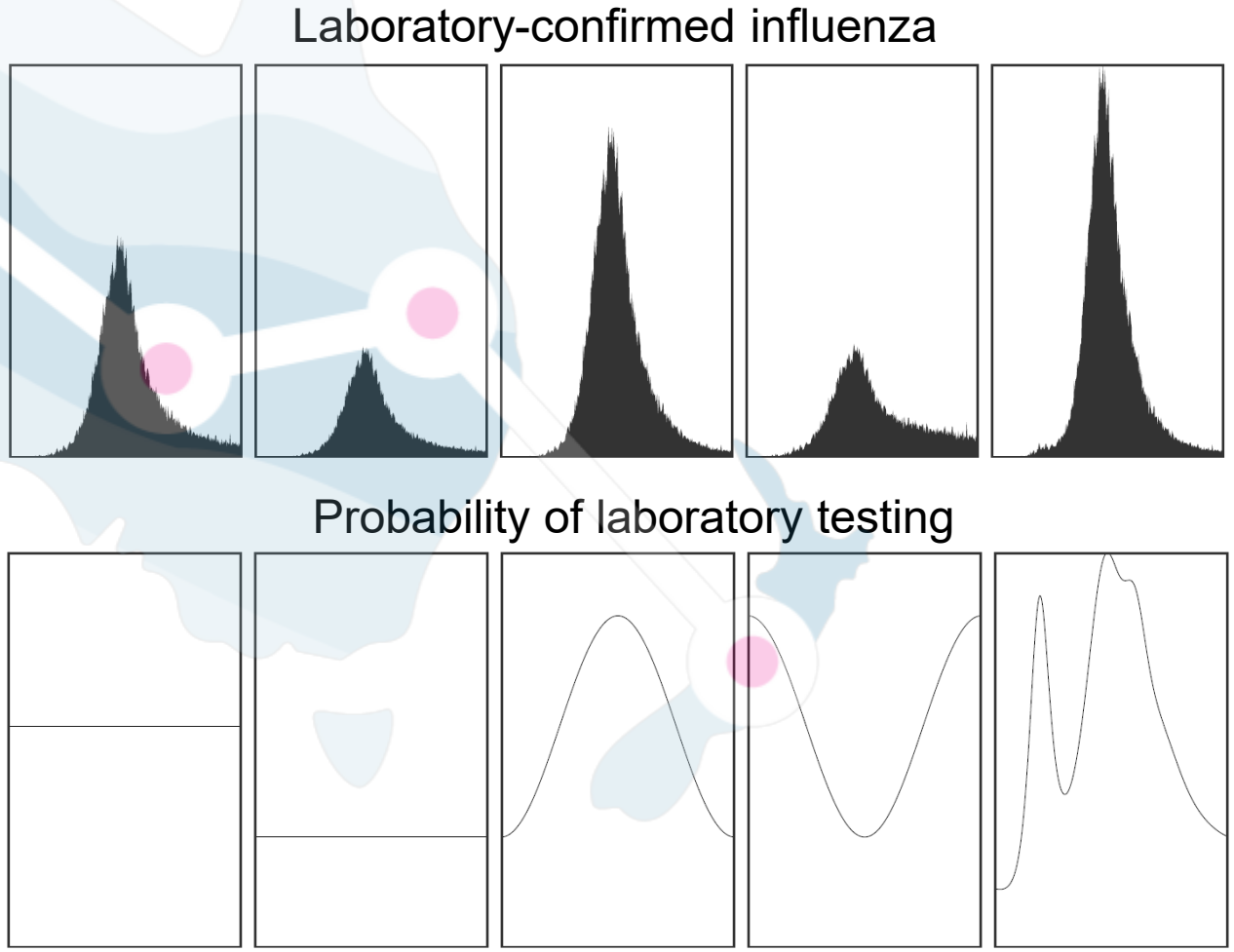




Eales et al. 2024. Influenza and Other Respiratory Viruses

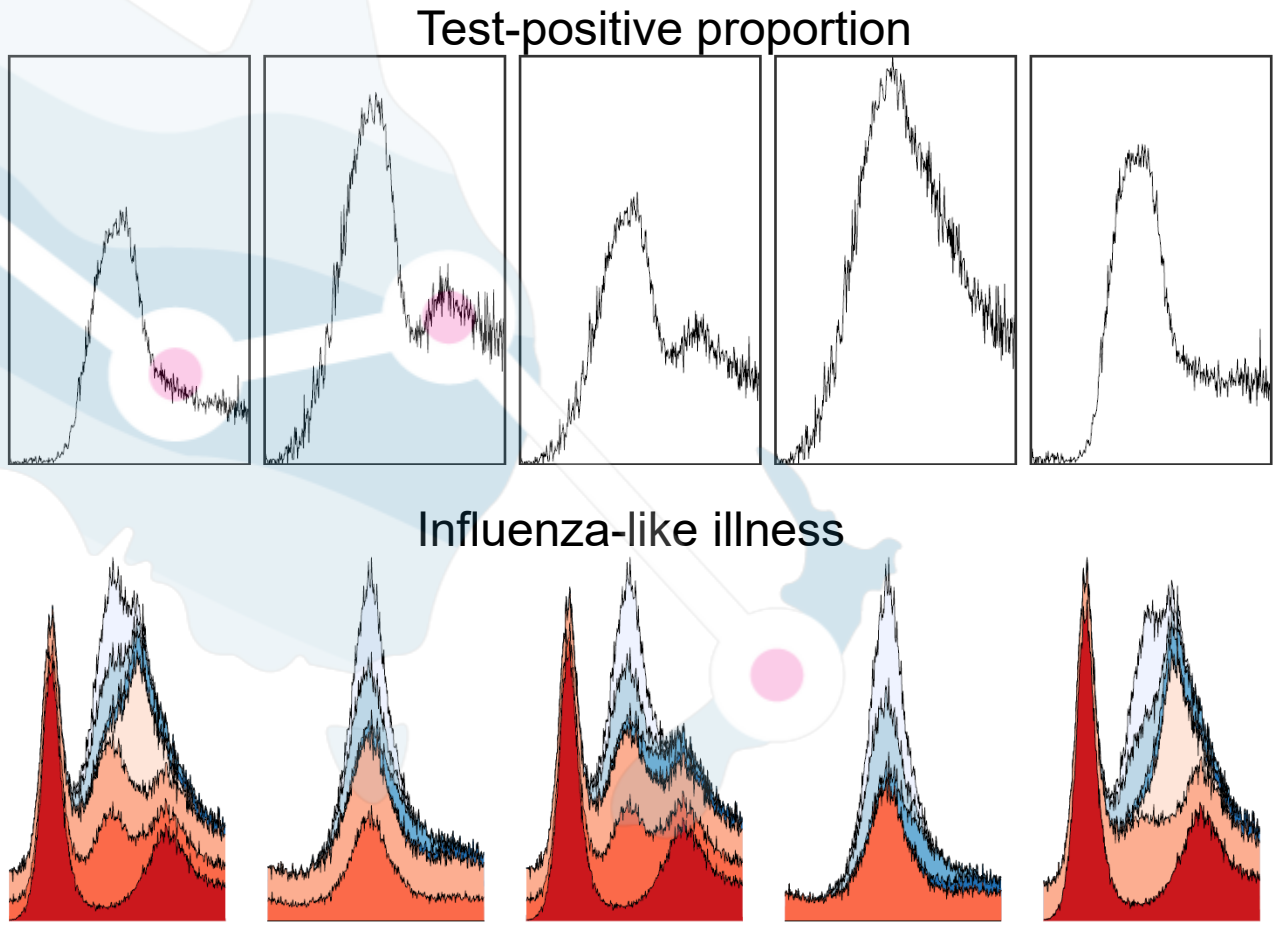
Surveillance indicators: Laboratory confirmed influenza (LCI)

- The rate of laboratory-confirmed influenza infections per unit time
- In some circumstances will not depend on other circulating pathogens
- Influenced by the probability of laboratory testing being performed given presentation to healthcare with ILI
- Probability of laboratory testing being performed can vary over time (both between seasons and within a single season)



Surveillance indicators: the test-positive proportion (TPP)

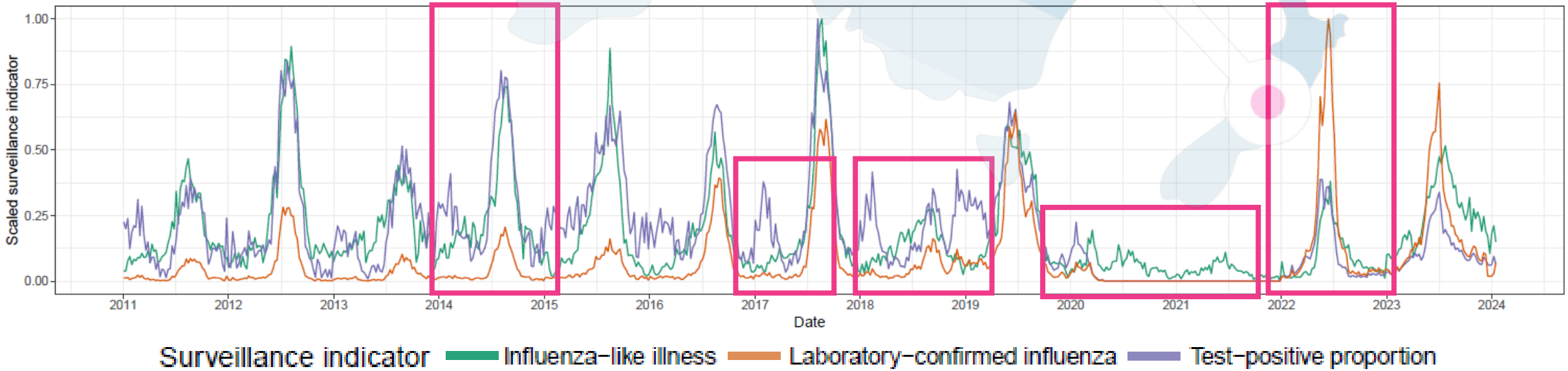
- The proportion of laboratory tests positive for influenza ('percentage positive')
- Most difficult to interpret surveillance indicator used to represent influenza infection incidence.
- Increases can reflect increases in influenza incidence or decreases in the incidence of other pathogens (e.g., SARS-CoV-2)
- If ILI was only caused by influenza the test-positive proportion would be constant



Comparison of influenza surveillance indicators

- Often substantial differences between the trends of these indicators within and across influenza seasons. For example:
  - Differences in relative peak magnitude
  - Differences in time of season ‘onset’

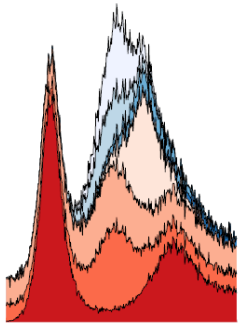
- So which indicator would be best to report?



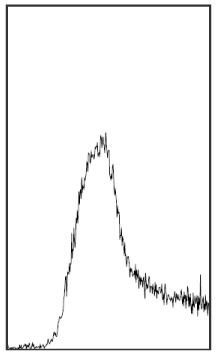
Report ILI+ as the primary surveillance indicator

- All three surveillance indicators (ILI, LCI, TPP) are dependent on multiple unknown and unmeasured quantities
  - Other circulating ILI-causing pathogens
  - Laboratory testing rates
- The data can be used to construct an indicator that more closely reflects trends in the underlying infection incidence
- ILI+: the rate of influenza-like illness multiplied by the test-positive proportion
- If laboratory testing also identifies other pathogens, then analogous surveillance indicators can be constructed for each pathogen

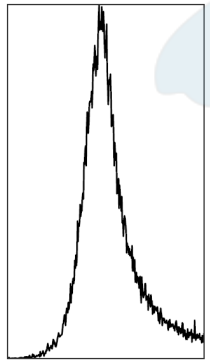
Influenza-like illness



Test-positive proportion

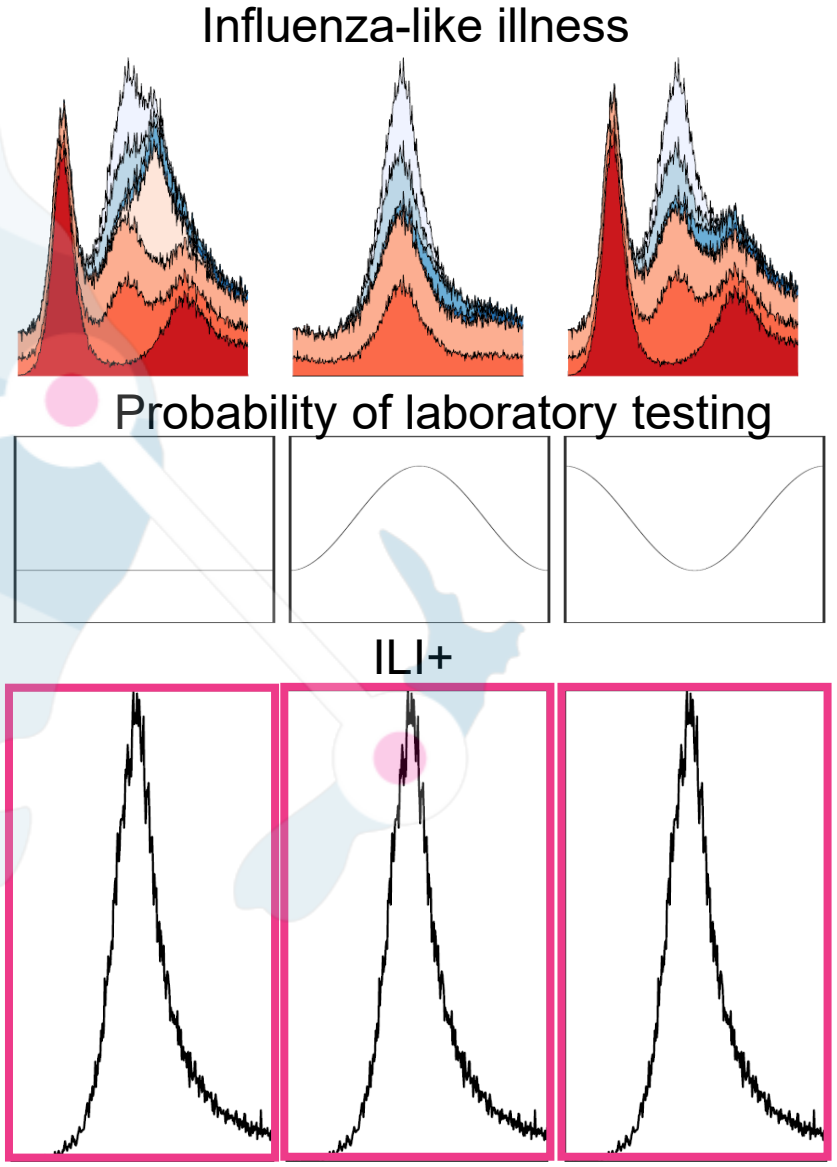


ILI+



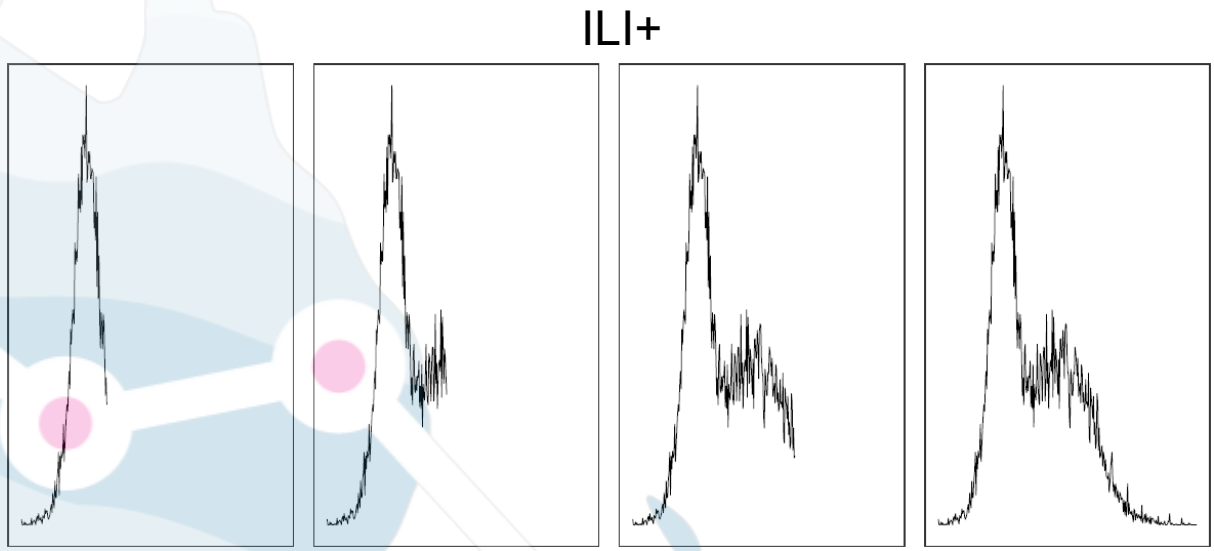
Report ILI+ as the primary surveillance indicator

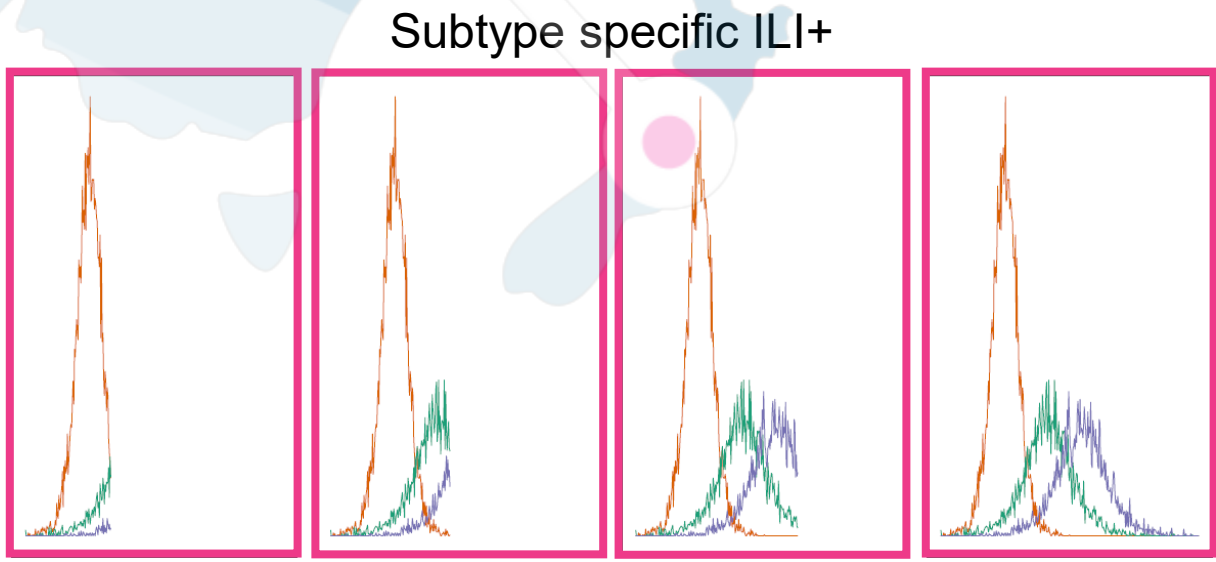
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- If laboratory testing also identifies other pathogens, then analogous surveillance indicators can be constructed for each pathogen



Stratify ILI+ by influenza type and subtype

- ILI+ can be stratified when there is subtype data available for a sample of laboratory-confirmed influenza
- Influenza types/sub-types exhibit different epidemic dynamics
- Making this distinction is important for predicting future epidemic activity

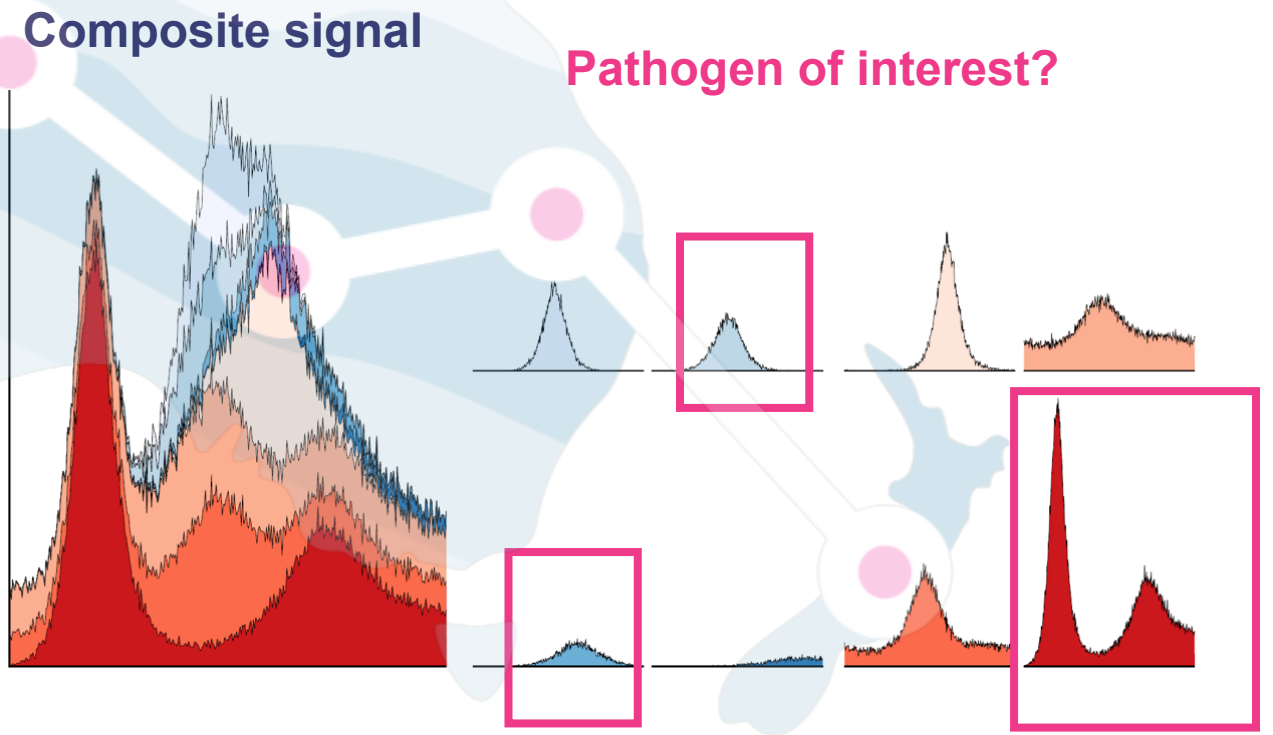




Relevance to other types of pathogen surveillance

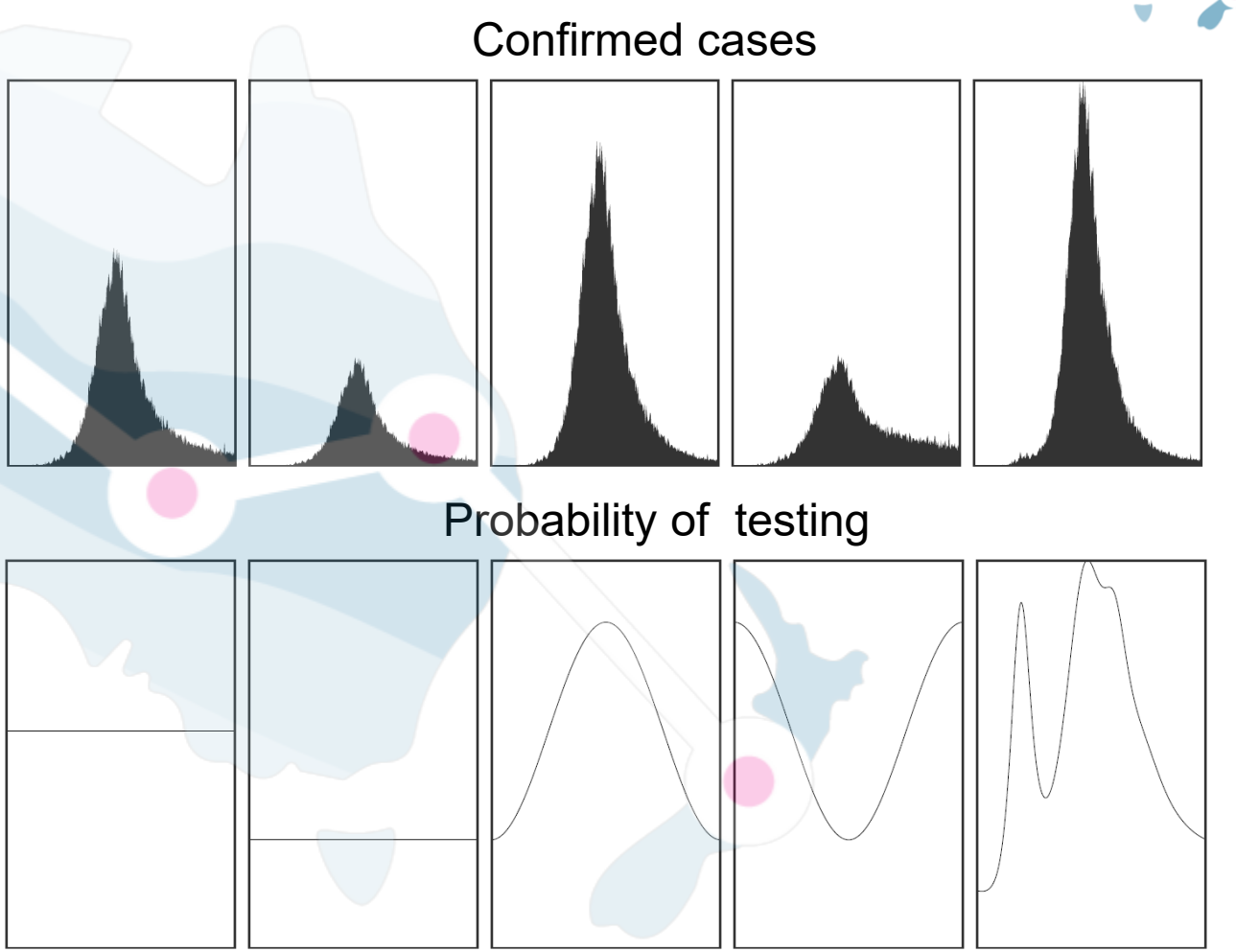
Syndromic surveillance

- Syndromic surveillance indicators routinely collected to monitor trends are often a composite of multiple pathogens
- For example, ILI and SARI contain contributions from multiple respiratory pathogens
- Trends in the overall syndromic signal may not be the same as trends in the pathogen of interest



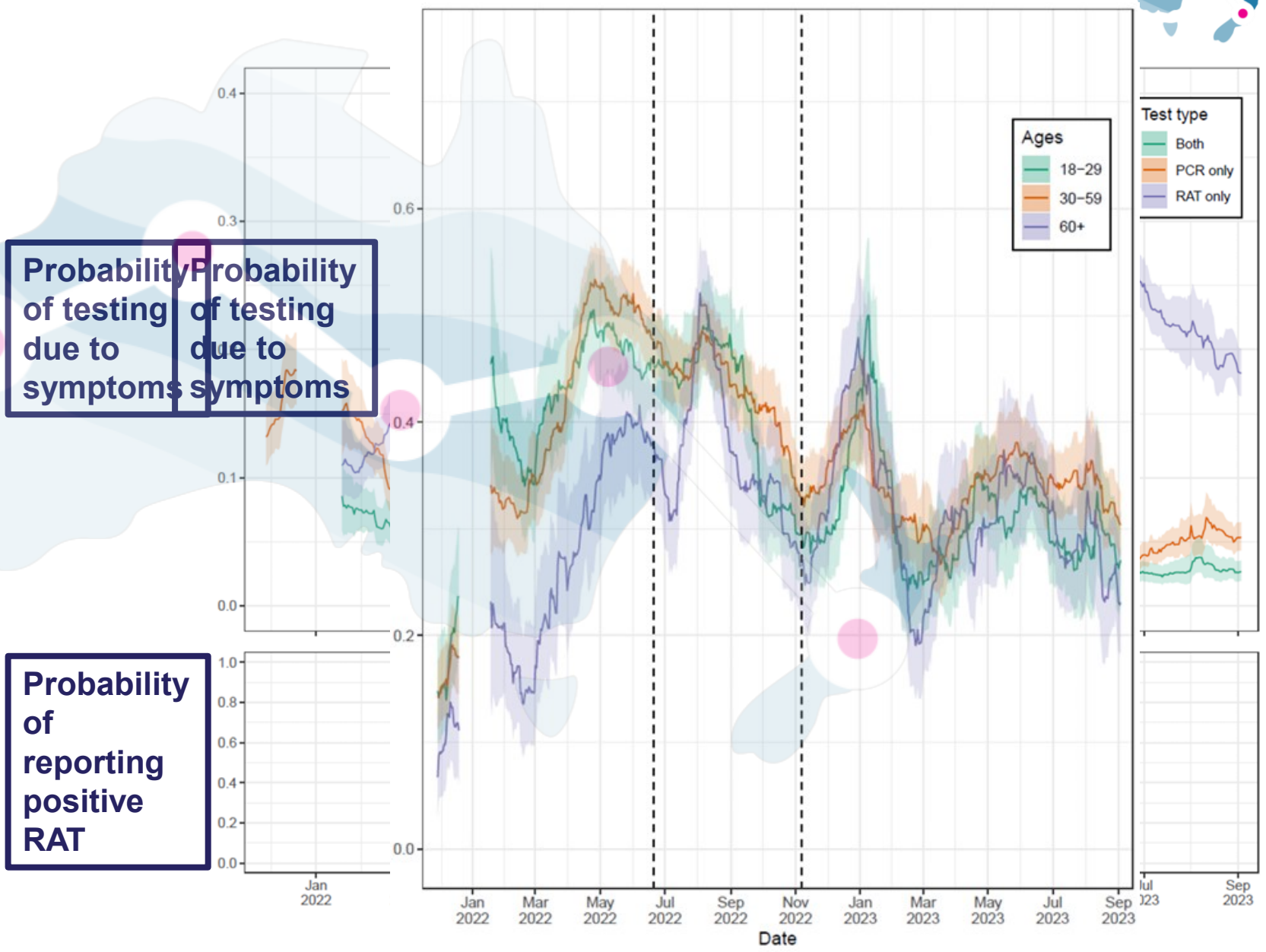
Confirmed cases (i.e. case-based surveillance)

- The rate of confirmed cases per unit time
  - Laboratory-confirmed
  - RAT-confirmed
- In some circumstances will not depend on other circulating pathogens
- Influenced by the probability of testing/reporting. Depends on:
  - Test-seeking behaviour (for each test type)
  - Test-reporting behaviour
  - Test availability



Confirmed cases (i.e. case-based surveillance)

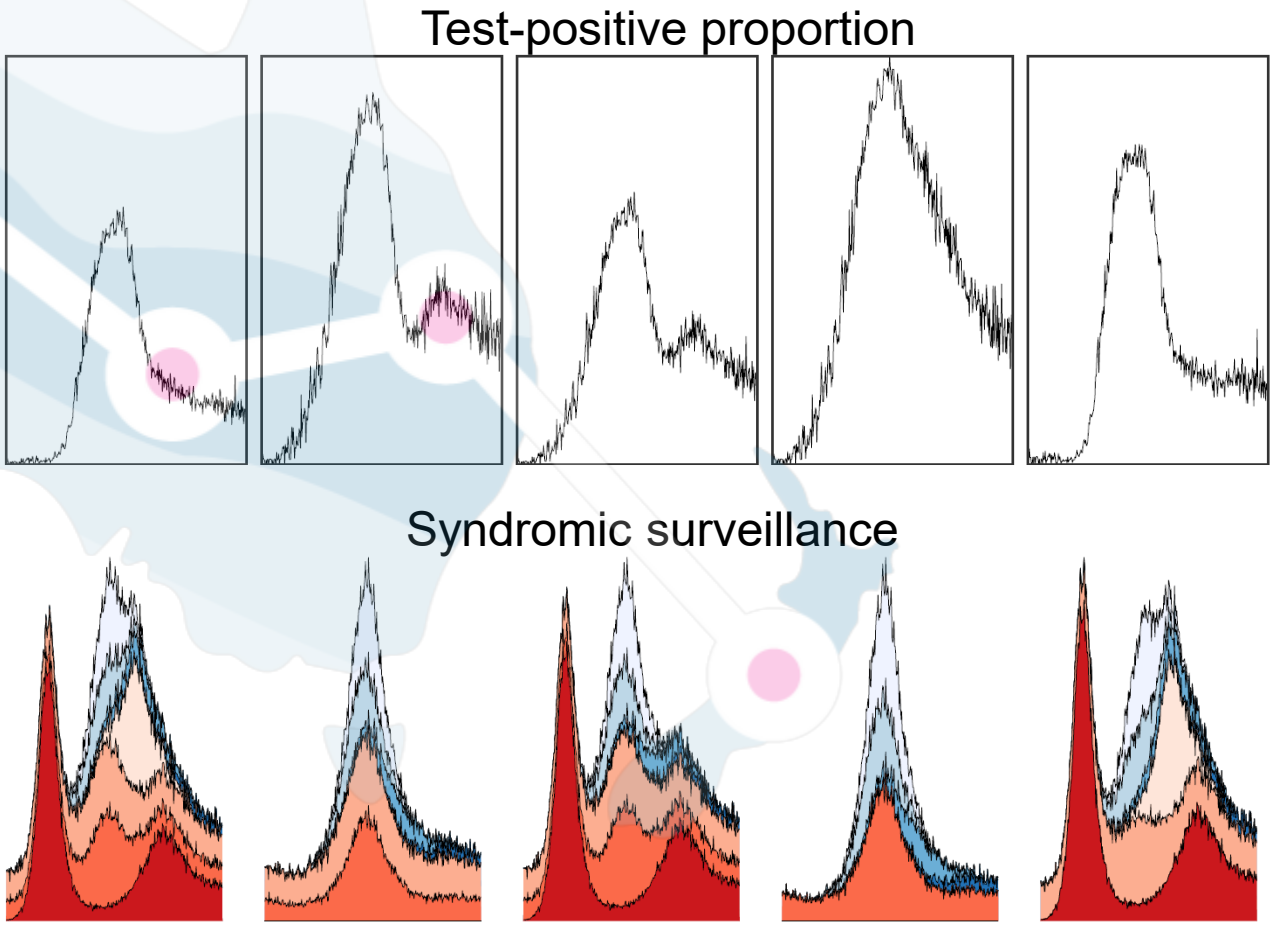
- The rate of confirmed cases per unit time is influenced by the probability of testing/reporting which can change over time



Eales et al. 2025. Mathematics in Medical and Life Sciences

Test-positive proportion

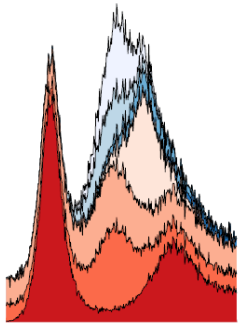
- The proportion of tests positive for a pathogen (‘percentage positive’)
- Difficult to interpret surveillance indicator
- Increases can reflect:
  - increases in the incidence of the pathogen of interest
  - decreases in the incidence of other pathogens



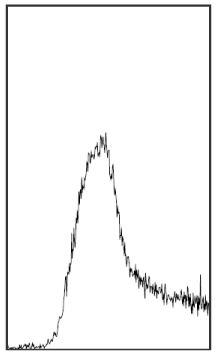
Combining syndromic surveillance with test-positive proportion

- Data can be used to construct a combined surveillance indicator that more closely reflects trends in the underlying infection incidence
- The rate of syndromic cases multiplied by the test-positive proportion
- If testing also identifies other pathogens, then analogous surveillance indicators can be constructed for each pathogen

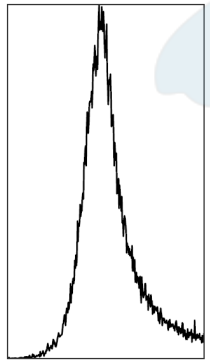
Syndromic surveillance



Test positive proportion

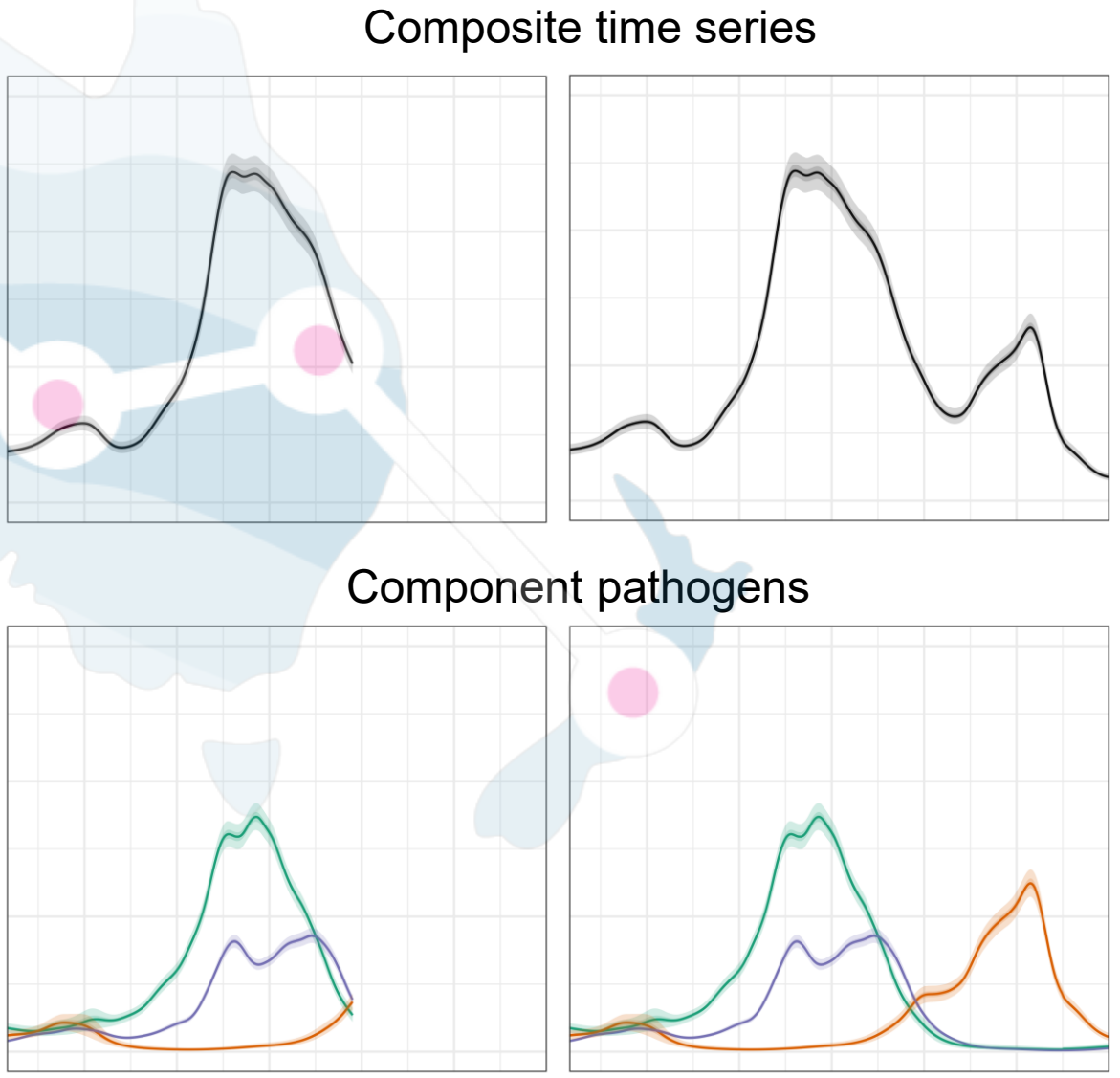


Combined surveillance indicator



Stratify surveillance indicators by component pathogens

- Many routine surveillance indicators are a composite of multiple pathogens:
  - Syndromic surveillance
  - SARS-CoV-2 variants
  - Influenza types/subtypes
  - Mpox clades
  - RSV groups
- Each component pathogen may exhibit different epidemic dynamics
- Supplementary data is often collected on the component pathogens contributing to the composite time series:
  - SARS-CoV-2 variants sequenced
  - Influenza typing/subtyping
  - Multiplex testing for respiratory pathogens

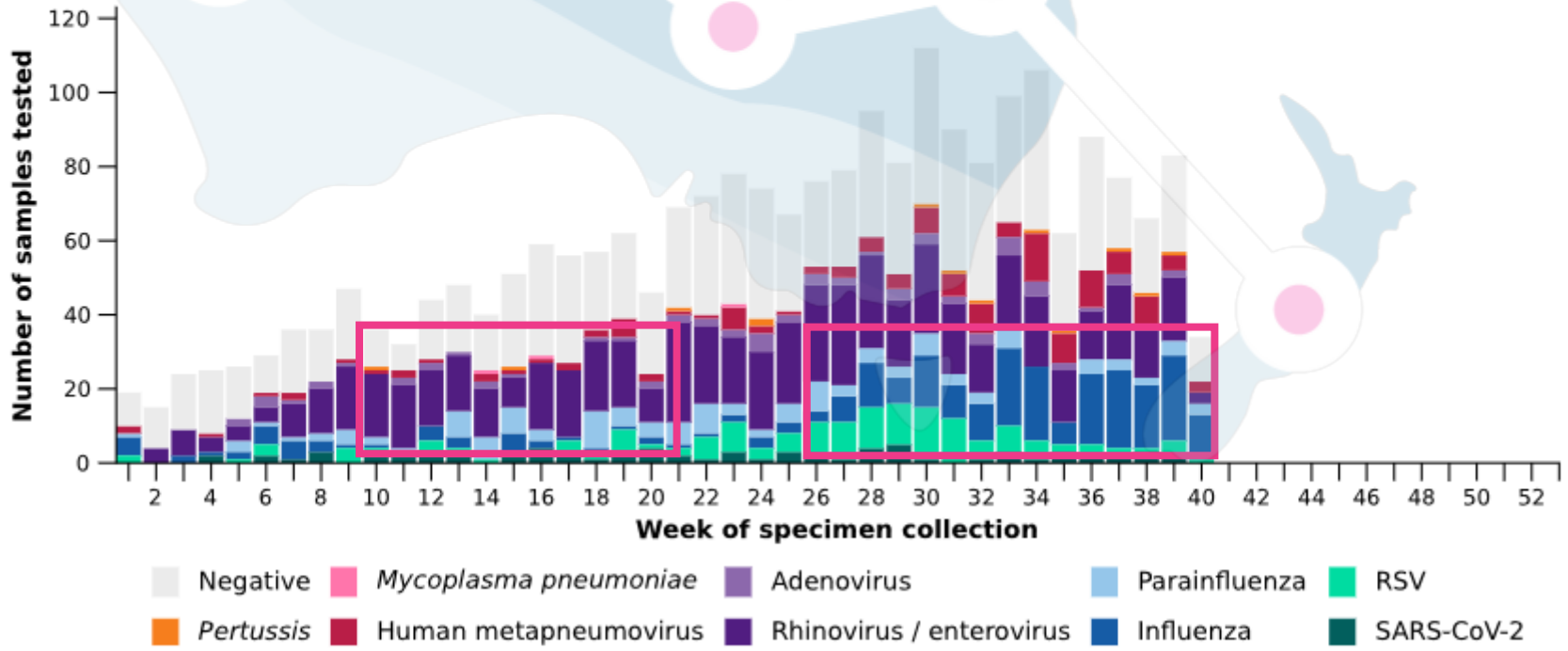


How are component pathogen data typically presented?

How are multi-component data typically presented?

- Is the proportion of ILI attributable to rhinovirus increasing or decreasing?
- Are ILI cases attributable to influenza increasing or decreasing?

Figure 14: Number of samples tested for respiratory pathogens among people with influenza-like illness attending sentinel general practice sites by respiratory pathogen and week of specimen collection, Australia, 1 January to 4 October 2026



Australian Respiratory Surveillance Report 21 September to 4 October 2026

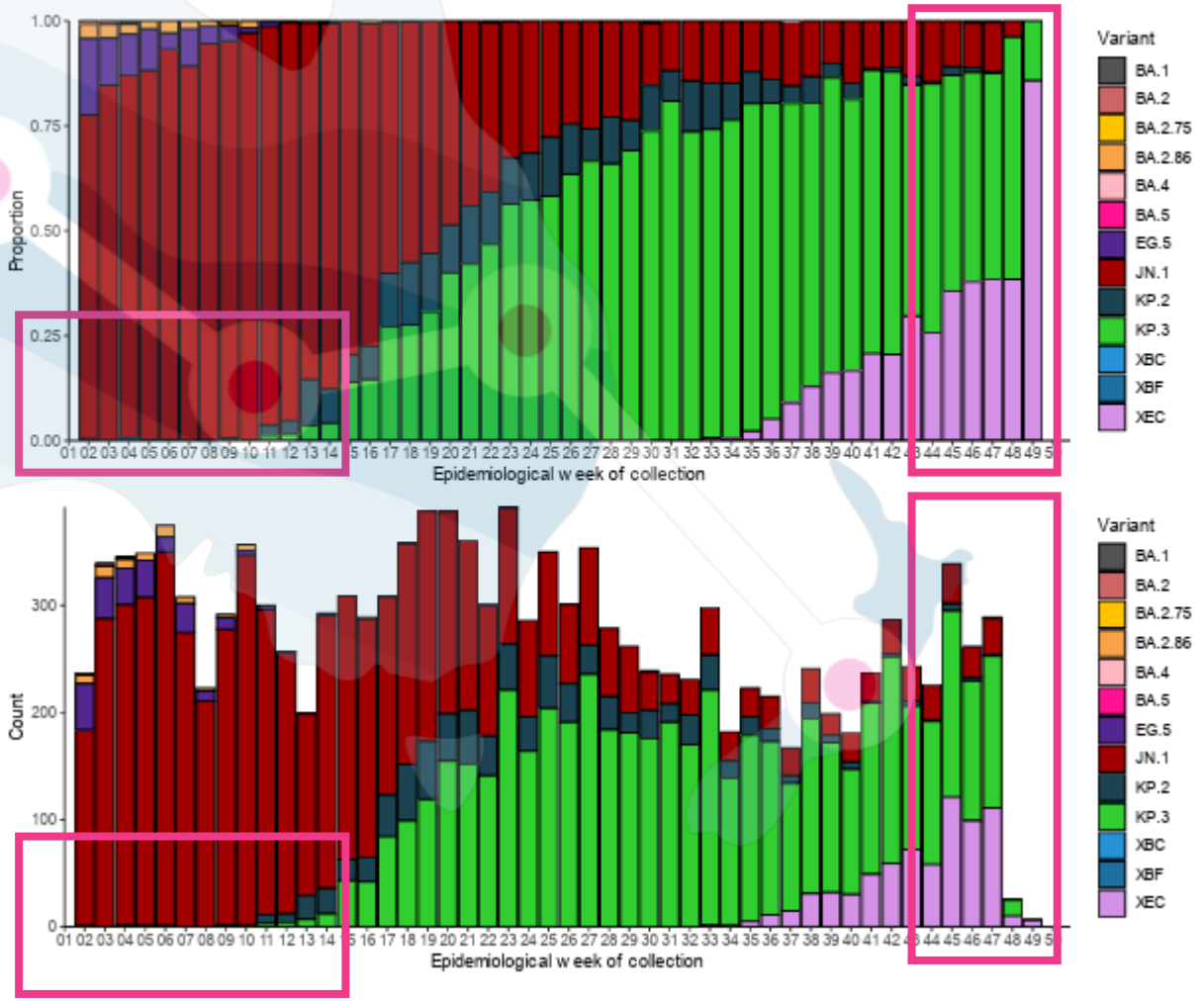
Figure 14

How is data currently presented?

- Which Omicron sub-lineage will be dominant in 2 months time?
  - KP.2
  - KP.3
  - JN.1

- What was the impact of the emergence of XEC on infection levels?

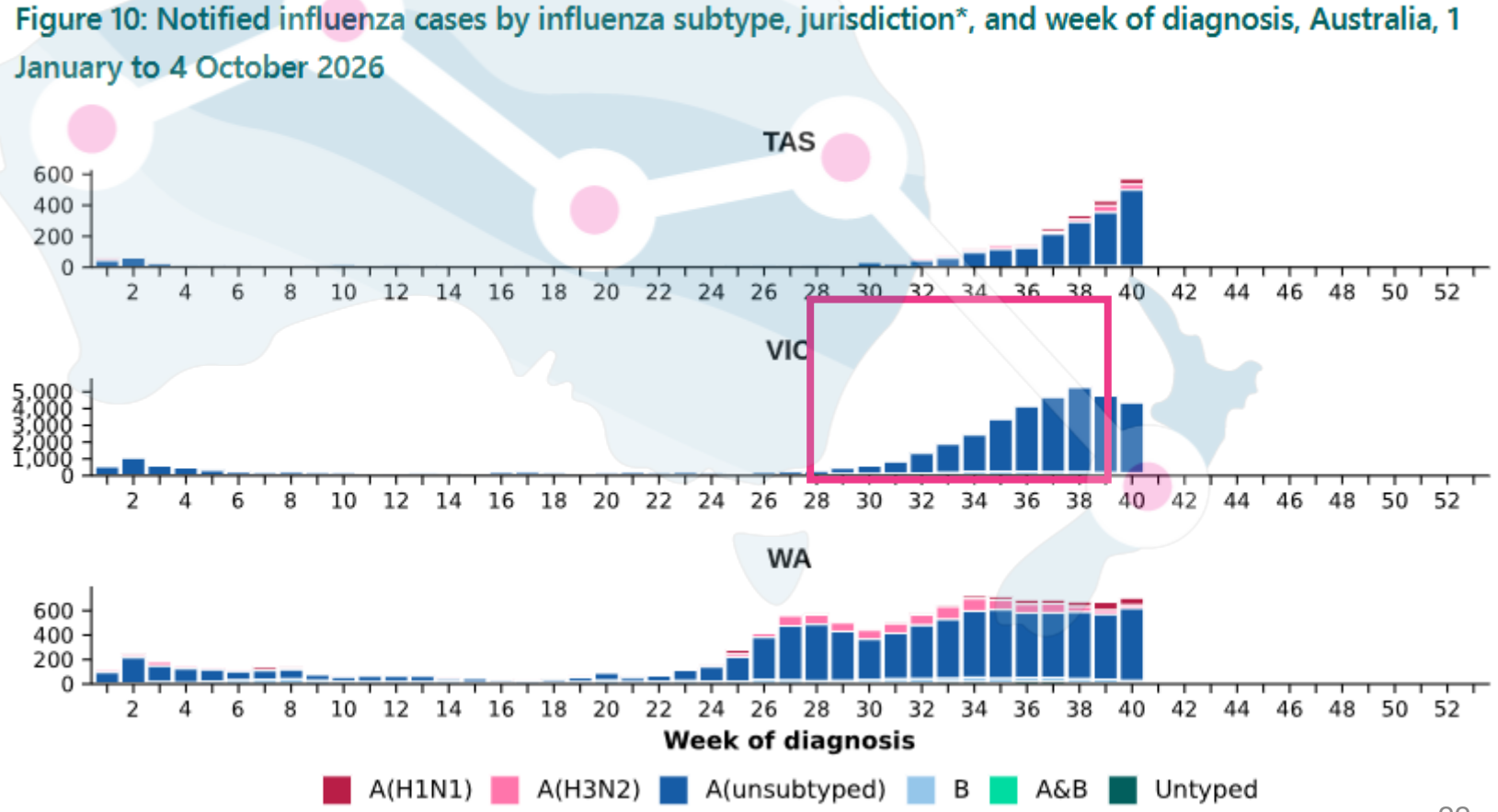
Figure 18: Omicron sub-lineage*^ sequences in AusTrakka by sample collection date, showing (A) proportions and (B) count per week^†, Australia, 1 January to 15 December 2024



Australian Respiratory Surveillance Report
Report 17, 2024
Section 5: Genomic surveillance and virology

How is data currently presented?

- Which influenza subtype dominated during the 2026 influenza season?
- Is influenza A H1N1 increasing or decreasing? How fast?

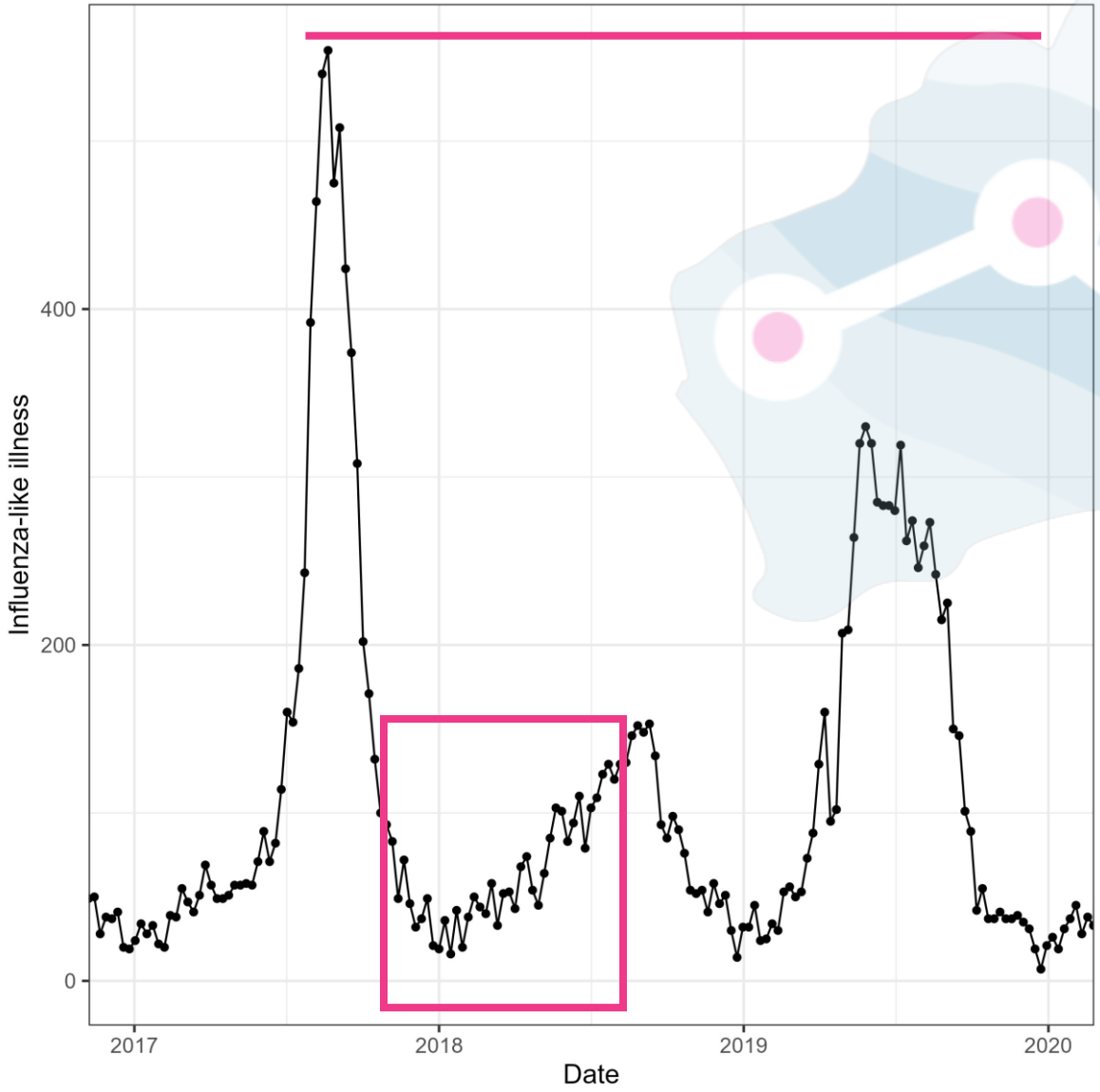


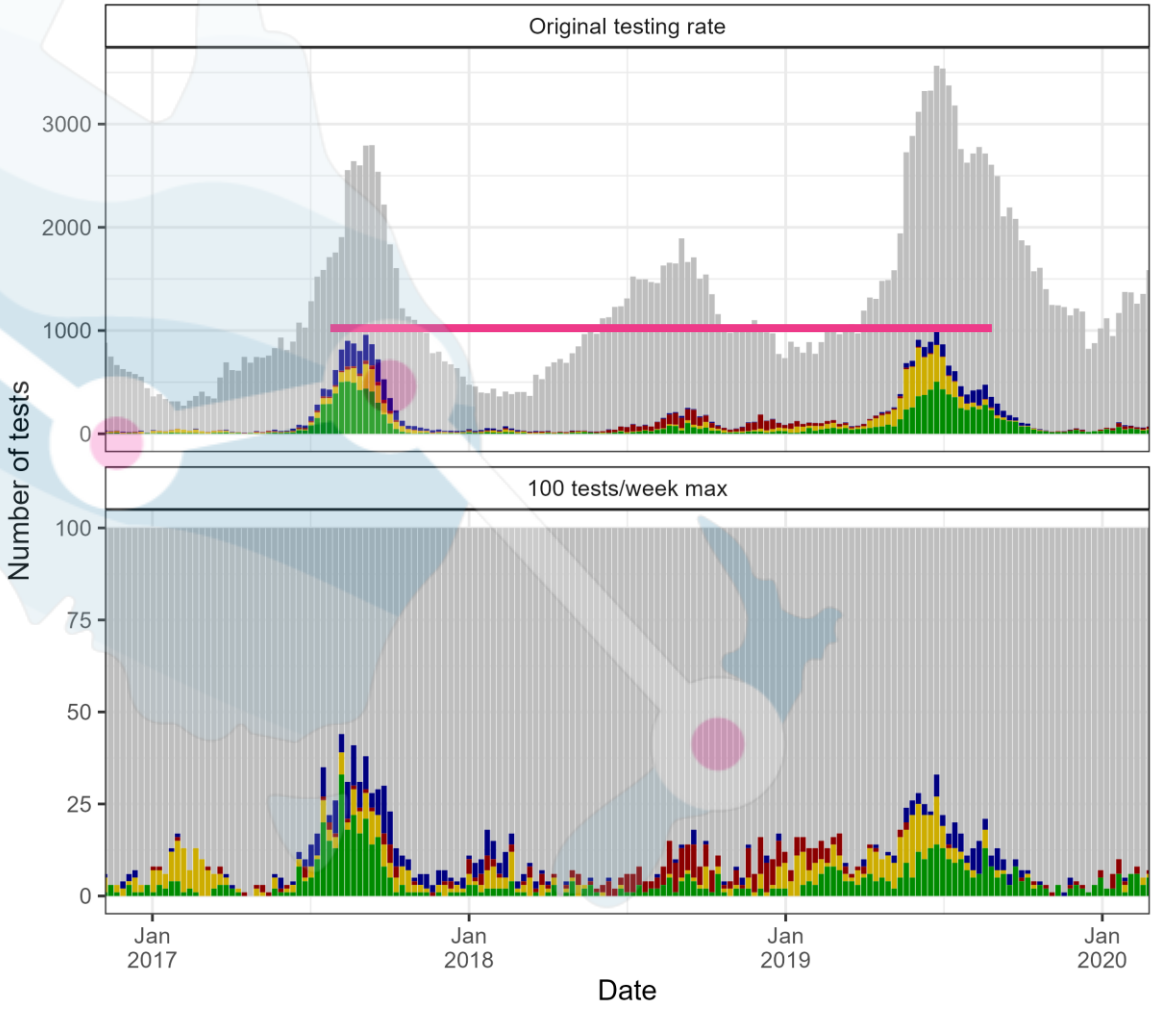
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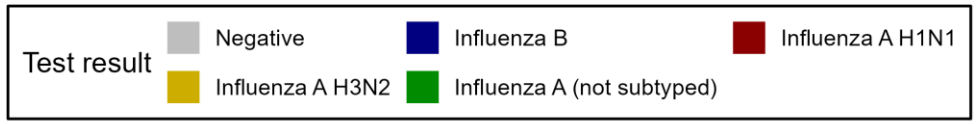
Figure 10

Statistical methods

What does ILI+ look like in practice?

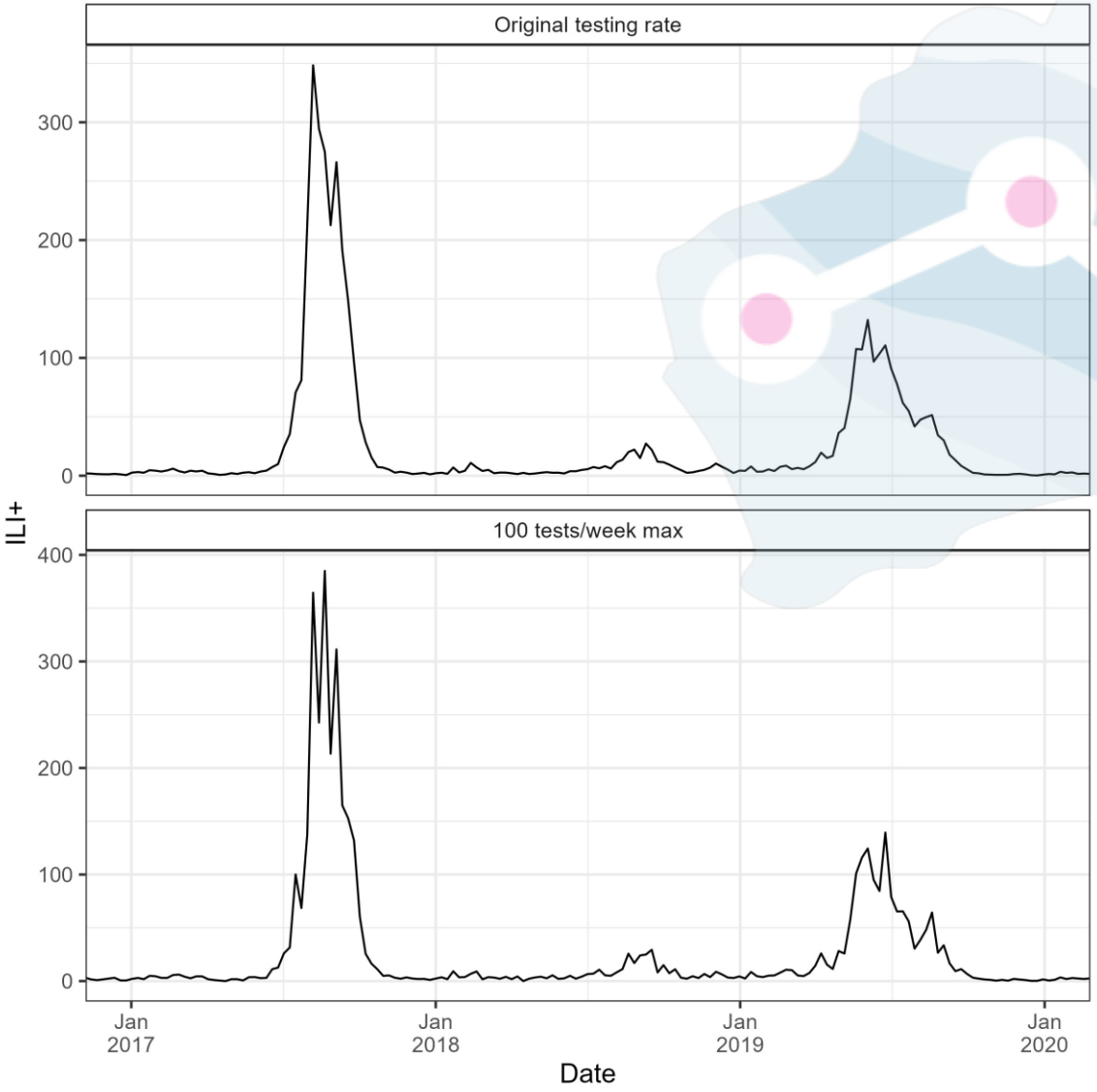


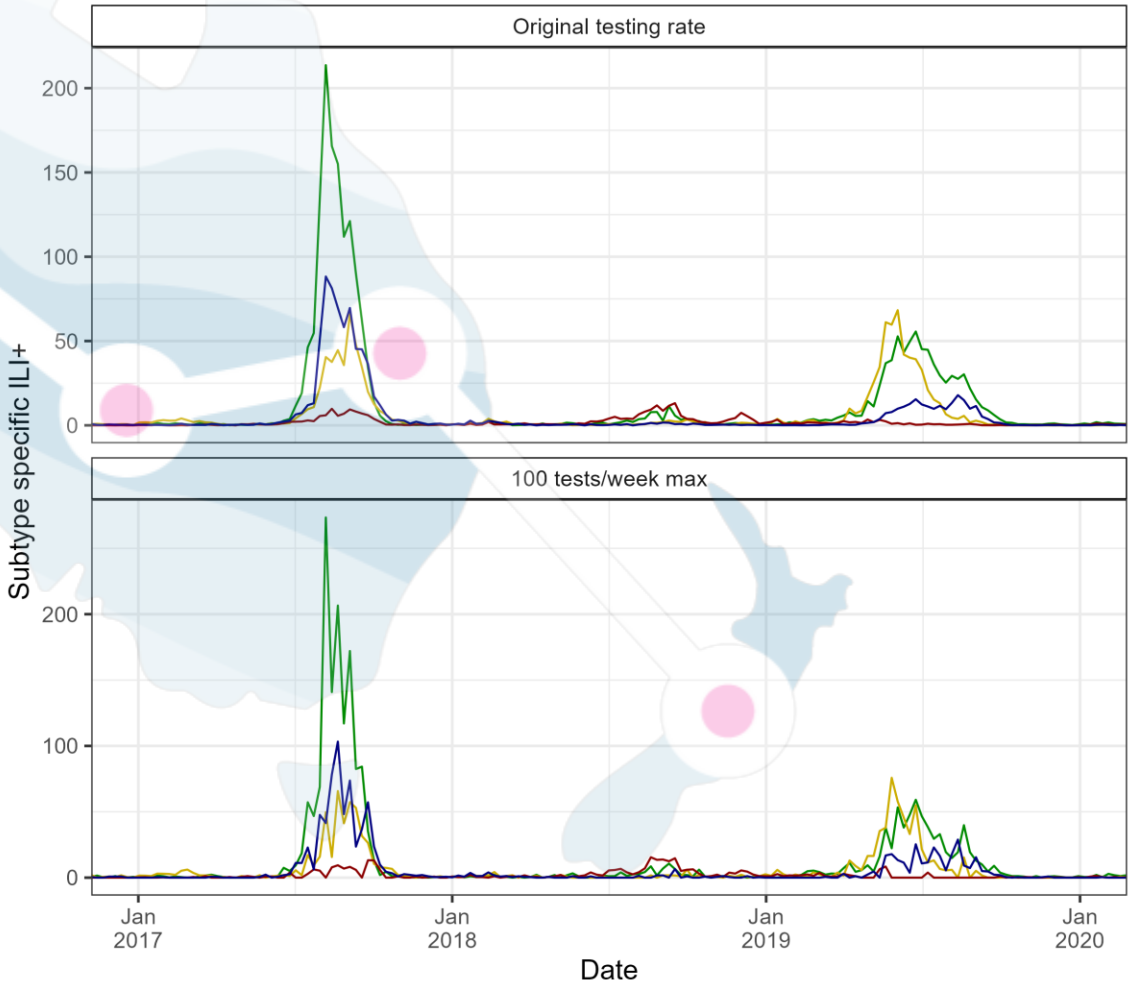




What does ILI+ look like in practice?

- Noisy and would be even noisier for daily data.



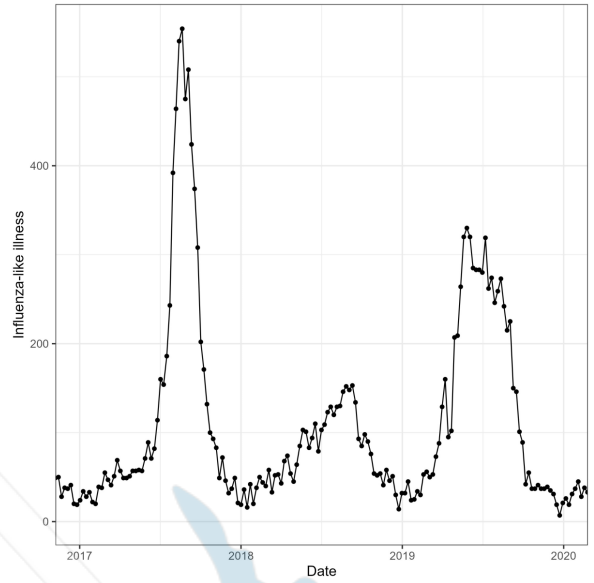


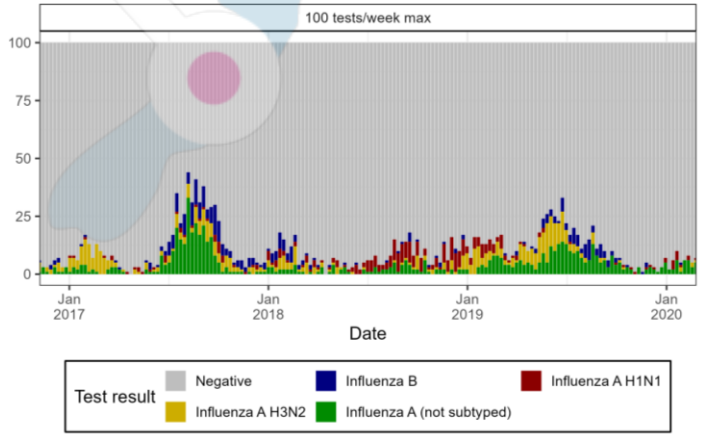
Statistical models can be used to account for noise

- Statistical model exists for simultaneously estimating epidemic dynamics of multiple pathogens using routinely collected surveillance data

- The model takes as input:
  1. A time series of count data (i.e. the composite time series)
  2. A time series of count data describing test results (i.e. the component time series).

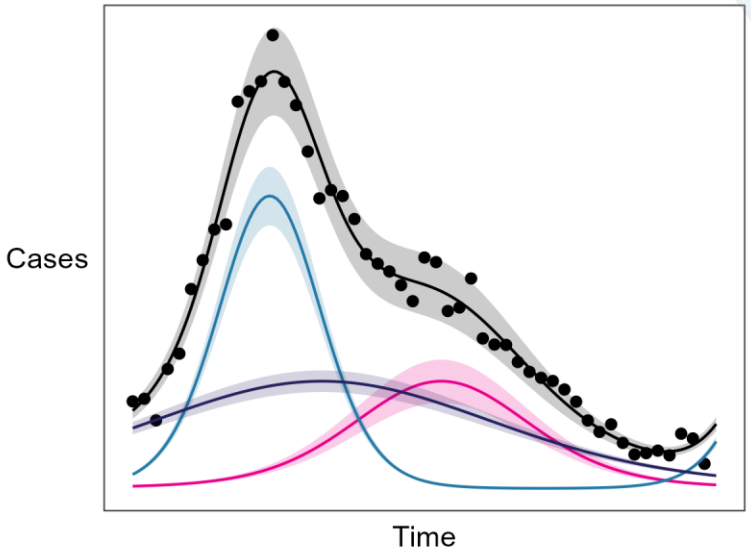
- Implementation of model in the R package ‘EpiStrainDynamics’

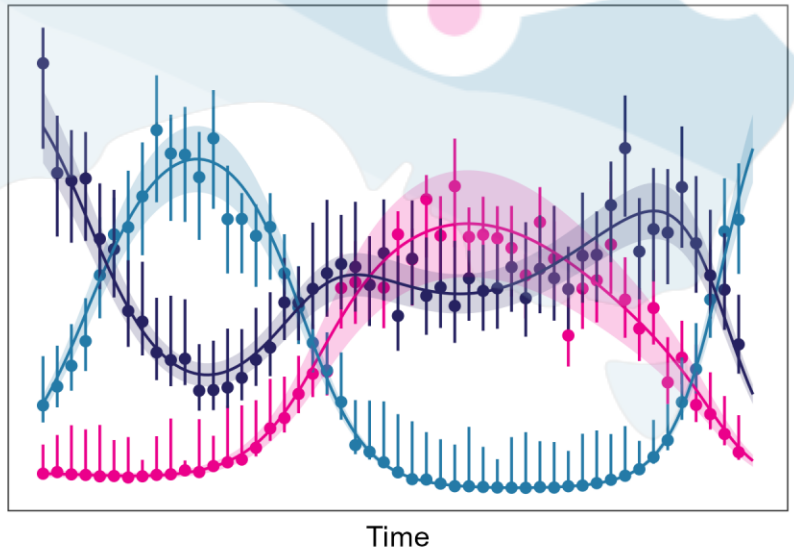


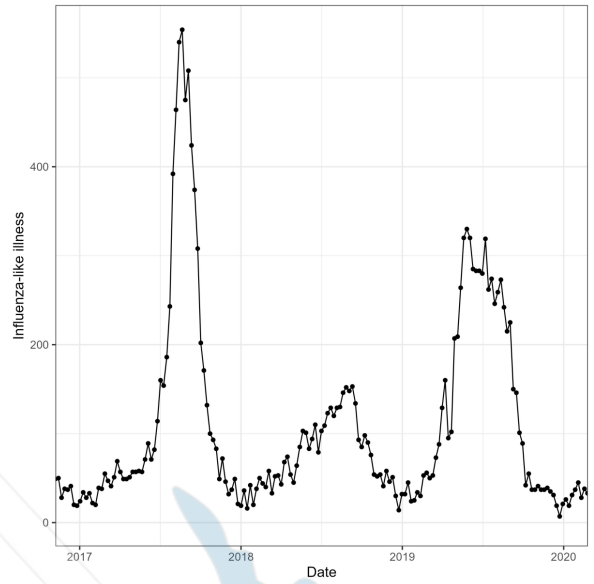


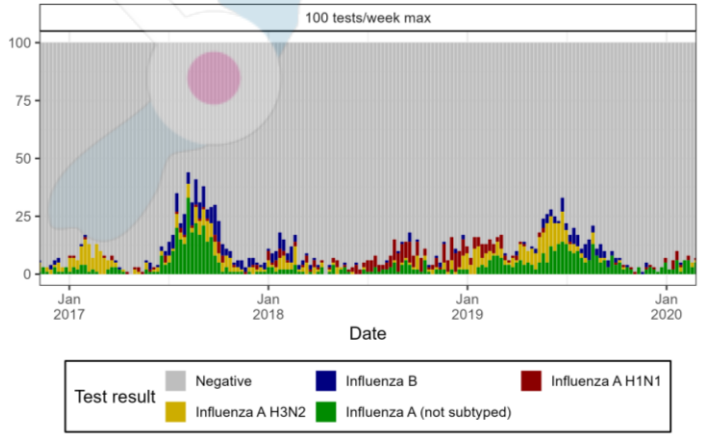
Statistical models can be used to account for noise

- Cases of each pathogen modelled as smooth functions
- Simultaneously fits:
  - Sum of modelled cases (for each pathogen) to composite time series
  - Relative proportion of modelled cases (for each pathogen) to component time-series



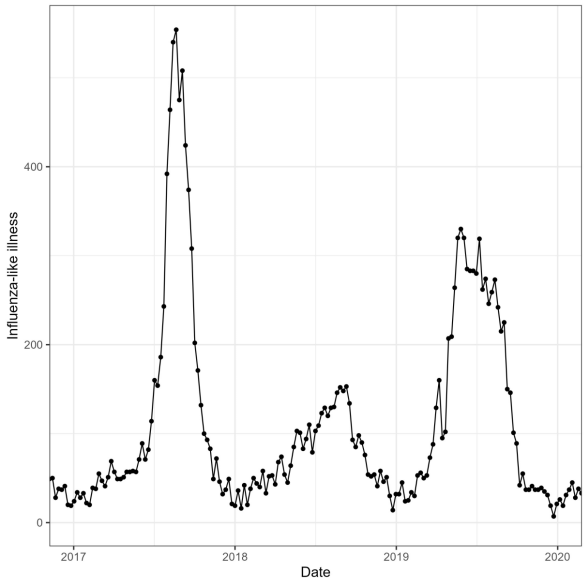


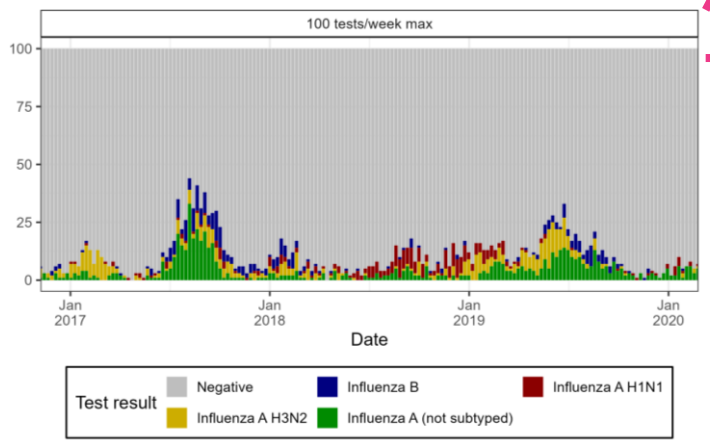


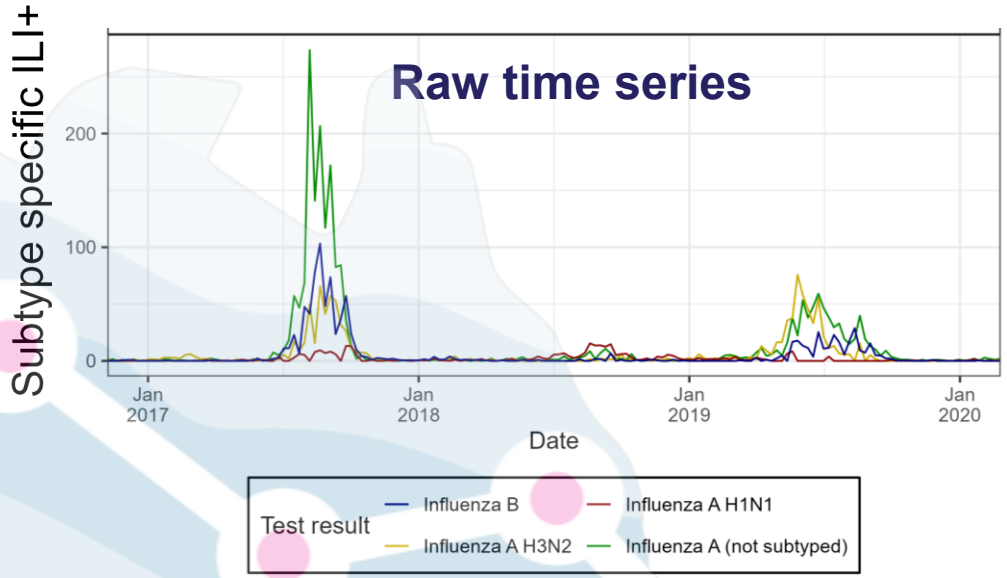


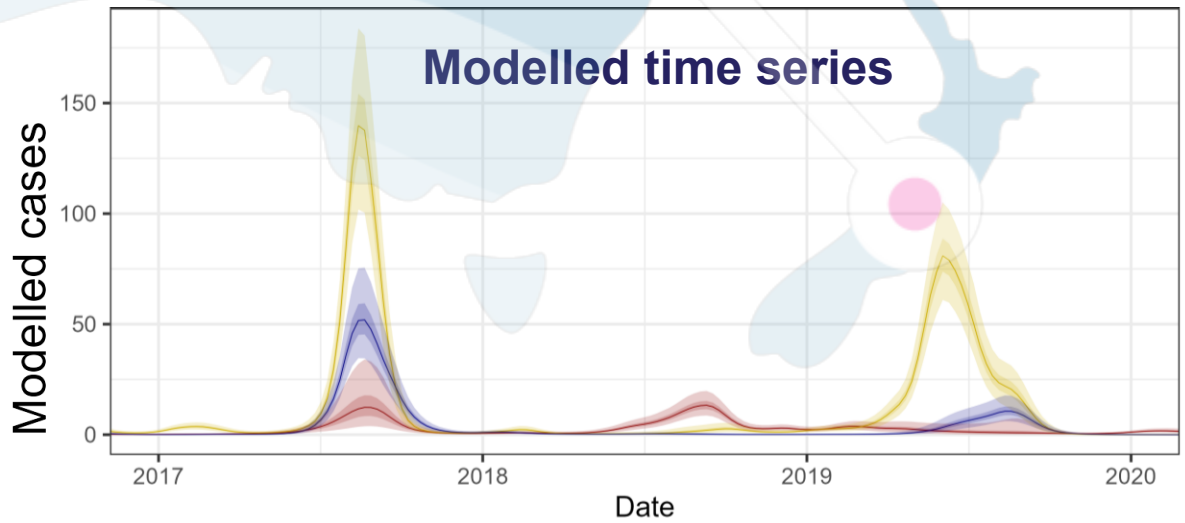
- For full methodological details please refer to Eales et al. 2025 American Journal of Epidemiology

Statistical models can be used to account for noise





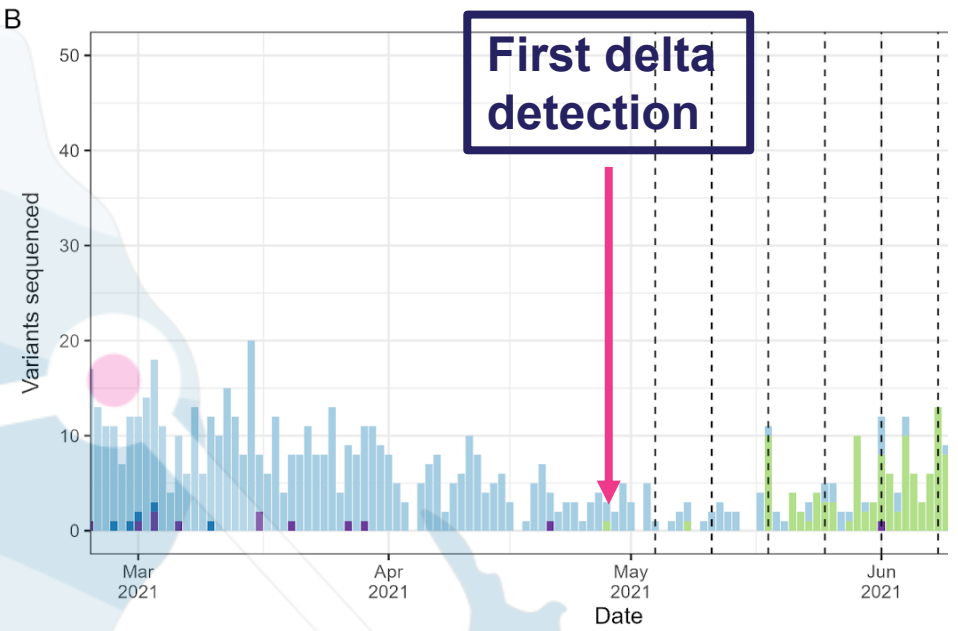


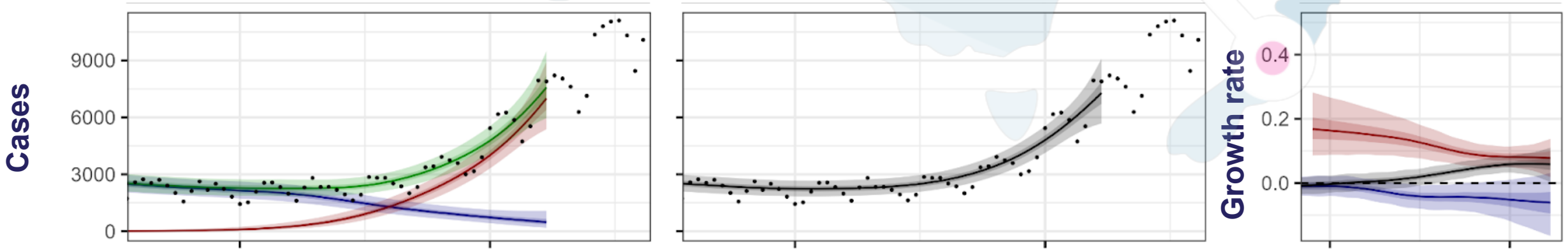


Case studies

Case study: Delta variant emergence in the United Kingdom

- Useful for real-time analysis
- Can identify components with higher growth rates
  - Even when composite time series is not increasing
- Not including variant data in analysis can delay identification of growth signal

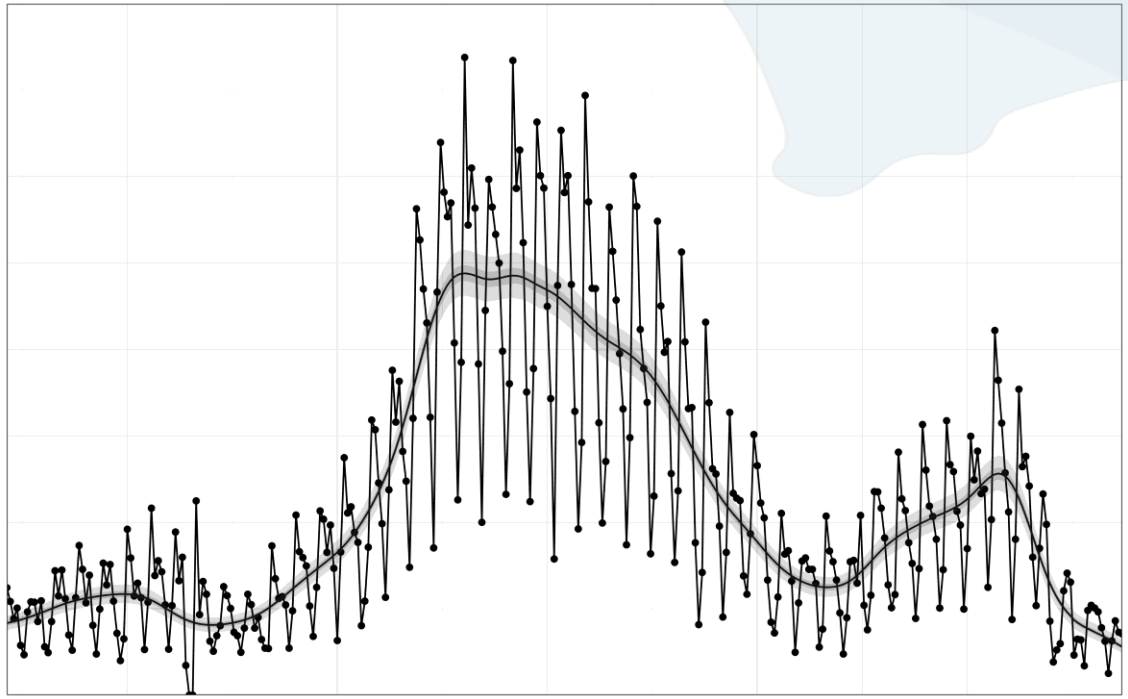


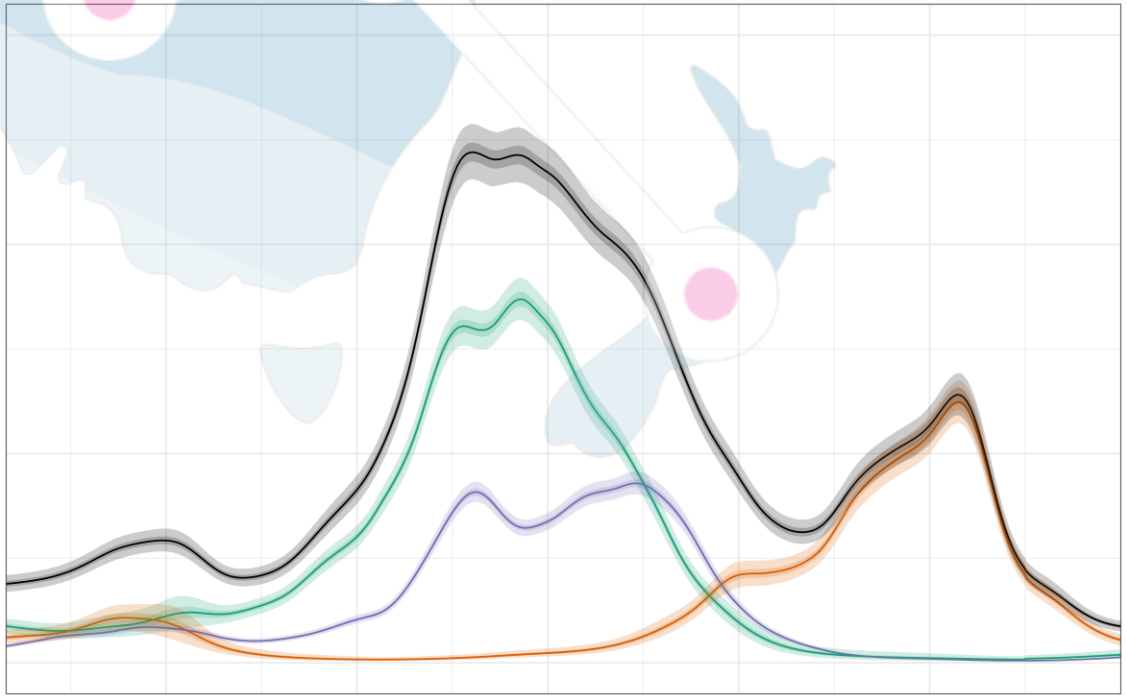


Eales et al. 2025 American Journal of Epidemiology

Case study: Real-time analysis of influenza subtypes in New South Wales 2025

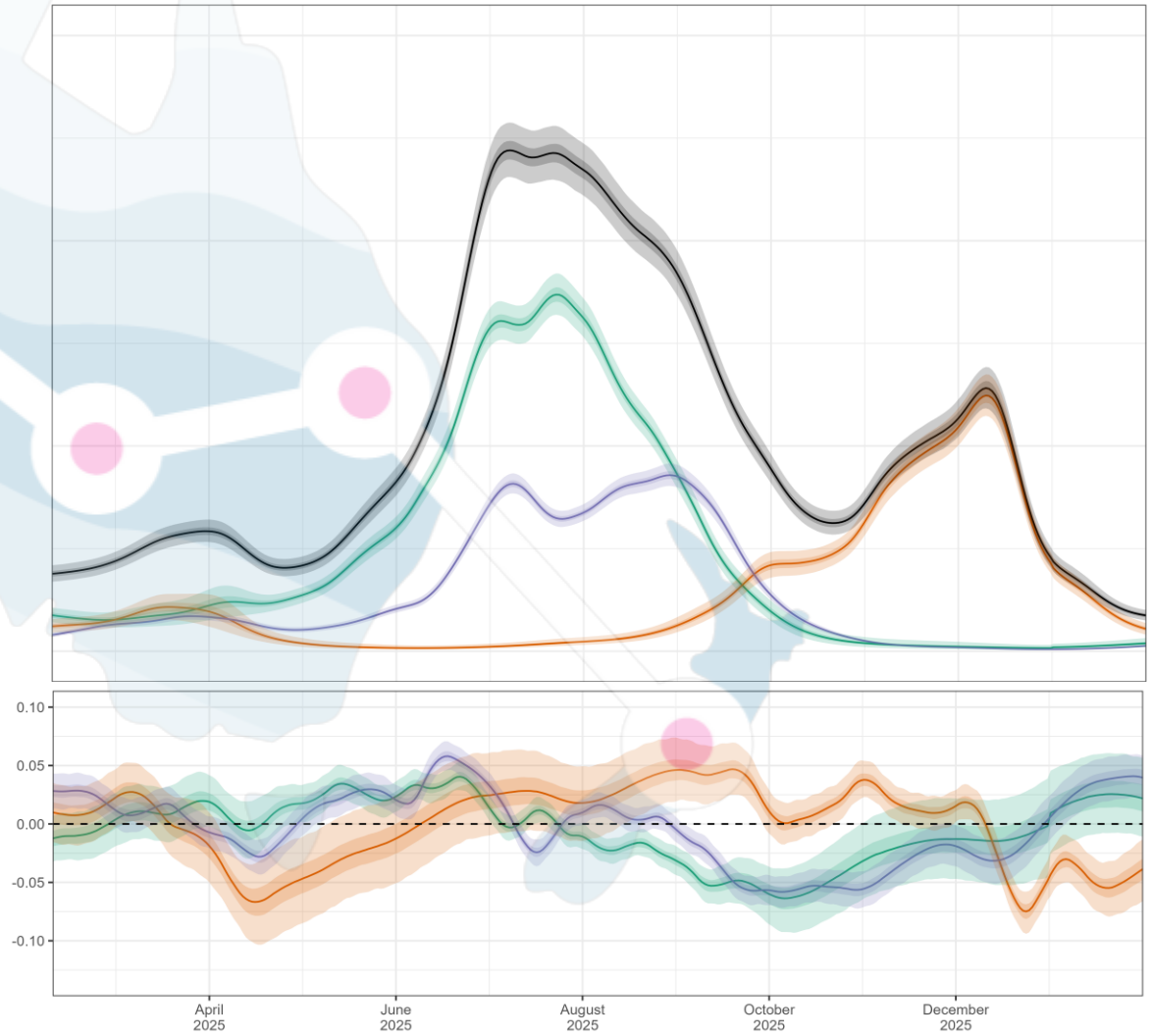
- During winter 2025 we performed analysis for laboratory confirmed influenza cases
  - Daily number of laboratory confirmed cases
    - Strong day-of-the-week effects
  - Results of typing/subtyping
    - Subtyping performed for ~2% of samples





Case study: Real-time analysis of influenza subtypes in New South Wales

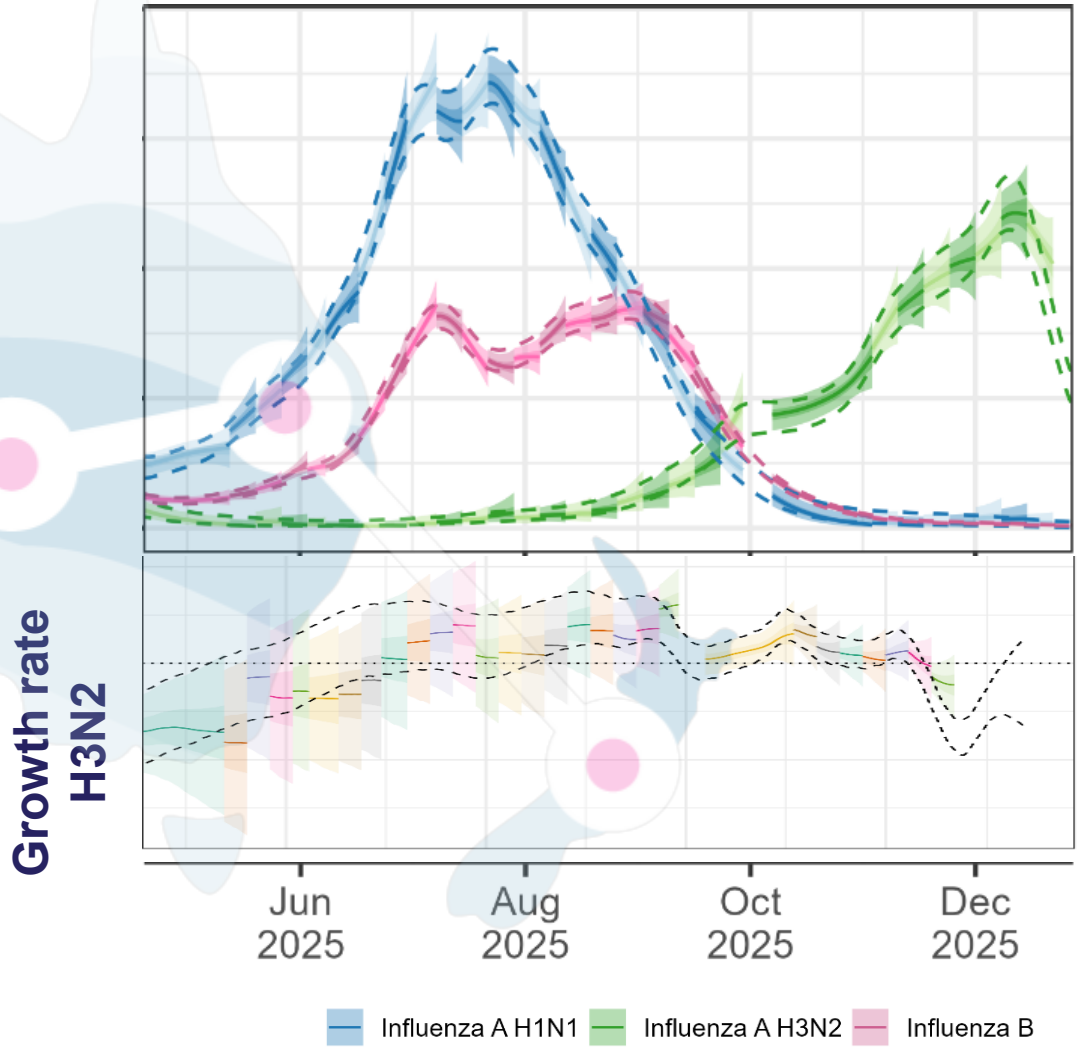
- Estimate smoothed trends in the:
  - Total number of influenza cases
  - Subtype specific influenza cases
  - Time-varying growth rates for each subtype
- Late H3N2 epidemic due to emergence of subclade K.
- Consistently positive growth rate from June onwards



Case study: Real-time analysis of influenza subtypes in New South Wales

- Real-time subtype specific estimates were broadly consistent with final end-of-season estimates
- Compare:
  - Solid lines and shaded regions (real-time estimates)
  - Dashed lines (95% CrI end-of-season estimates)

- Real-time evidence of growth from July-onwards



Eales et al. 2026. medrxiv

Summary



- We have discussed some of the biases present in epidemic time series that are routinely measured and reported
  - For a more extensive description see the further reading for this lecture
- We have demonstrated that epidemic time series data are often composed of multiple signals from different pathogens. For example:
  - SARS-CoV-2 variants
  - Other respiratory pathogens contributing to ILI
  - Mpox clades
  - Dengue serotypes
  - RSV groups
- Statistical models exist that can infer trends in multiple pathogens from routine surveillance data
  - R package is available (EpiStrainDynamics)
  - Example code for running the model provided in course materials

References

- Eales O, McCaw JM, Shearer FM. Biases in routine influenza surveillance indicators used to monitor infection incidence and recommendations for improvement. Influenza Other Respi Viruses. 2024;18: e70050.
- Eales O, McCaw JM, Shearer FM. Challenges in the case-based surveillance of infectious diseases. R Soc Open Sci. 2024;11: 240202.
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- Eales O, et al. Temporal analysis of respiratory virus epidemics in Victoria over winter 2024. Communicable Diseases Intelligence. 2025 [in press]